

AAGAM PARAG SHAH

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EDUCATION

University of Illinois, Urbana-Champaign, USA

August 2025 - May 2027

Master of Science in Information Management

GPA: 3.89/4.0

- **Relevant Coursework:** Data Science, Information & Statistics, Data Warehousing & Visualizations

Manipal University, Jaipur, India

July 2019 - July 2023

Bachelor of Technology in Computer & Communication (Minors: Data Science)

CGPA: 8.89/10.0

- **Relevant Coursework:** Object Oriented Programming, Operating Systems, Data Structures and Algorithms, Computer Networks, Data Analysis, Machine Learning, Database Management

SKILLS

Programming / Scripting Languages: Python, SQL, C, C++, JavaScript, Bash, Perl, R

Data Engineering & Analytics: ETL, Data Modeling, dbt Core, Schema Design, Feature Engineering, DuckDB

Databases & Storage: PostgreSQL, MS SQL Server, Object Storage (S3-compatible / MinIO), Relational Databases

Workflow, Infrastructure & DevOps: Docker, Docker Compose, Git, GitHub, Linux, REST APIs, Kubernetes, Jenkins

ML & Data Science: Scikit-Learn, TensorFlow, Keras, Pandas, NumPy, NLP, LangChain, Model Deployment

Visualization & BI Tools: Tableau, Power BI, BusinessObjects, Streamlit, Splunk, Jupyter

WORK EXPERIENCE

Cisco Systems, Bangalore, India

Aug 2023 - July 2025

Technical Consulting Engineer - Network Security

- Engineered automated data processing pipelines analyzing **500+** daily production events from distributed **logging systems (Splunk)**, reducing pipeline failures **60%** through **ETL workflows** and data quality validation.
- Architected production ML system for **automated packet analysis** achieving **80%** classification **accuracy** on **10K+** daily captures using Python, PyShark, **Langchain**, and **NLP models**; implemented end-to-end pipeline with data preprocessing, feature engineering, and real-time inference **API** via **Streamlit**.
- Built scalable log analysis infrastructure processing **100K+ network events** daily with data **cleaning**, **transformation**, and **aggregation** pipelines achieving **90% accuracy**: optimized query performance by **40%**.

Cisco Systems, Bangalore, India

Jan 2023 - July 2023

Technical Intern

- Developed an **Agentic AI** chatbot using **Python** and **ML** for document search and ticket routing, achieving 90% user satisfaction and **85%** classification accuracy.
- Implemented a **RAG (Retrieval-Augmented Generation)** pipeline to process 1K+ daily queries, utilizing NLP-based intent classification for **high-precision** information **retrieval**.

PROJECTS

COVID-19 Data Pipeline & Analytics

Jan 2023 - June 2023

Tech: Python (Pandas, NumPy), MS SQL Server, Tableau, ETL, Data Modeling, Star Schema

- Designed end-to-end ETL pipeline processing **1M+** records from multiple datasets (Our World in Data, WHO) with **automated** data ingestion, validation, and transformation using Python **Pandas**.
- Built dimensional data warehouse in **MySQL** with star schema supporting **20+** analytical queries; implemented data quality framework ensuring **99.5%** accuracy across **50+** countries.
- Created **Tableau** dashboards with **5+ visualizations** for mortality trends and vaccination analysis.

Real-Time ML Drowsiness Detection System

Sept 2021 - Dec 2021

Tech: Python, TensorFlow, Keras, OpenCV, Feature Engineering, Model Deployment

- Developed deep learning model achieving **90%** accuracy on **1,000+** images using **TensorFlow/Keras**; built **ML pipeline** with data preprocessing and **feature engineering** (facial landmarks, eye aspect ratio) as team lead.
- Implemented **data augmentation** increasing dataset by **300%**, reducing overfitting **25%**; designed **real-time** inference system processing 30 FPS.

Threat & Risk Analytics Platform (UIUC)

Tech: Python, Docker, Postgres, MinIO, dbt, DuckDB, Airflow

Ongoing

- Built a cloud-portable **MLOps** engineering platform using **Docker** Compose to implement **ELT** pipelines with automated quality checks via **dbt** tests.
- Designed a layered **Bronze/Staging/Marts** architecture for production-style data modeling, optimizing for **scalability** and future orchestration.
- Enabled fast local **OLAP** analysis using **DuckDB** to derive meaningful insights and perform data storytelling for risk assessment.

RESEARCH & LEADERSHIP

Metaheuristic Algorithms for Cloud Task Scheduling

Feb 2022 - July 2022

- Developed metaheuristic algorithms achieving **85%** efficiency with **80%** resource utilization to minimize resource consumption and improve system performance over existing methods.

Treasurer, MUJ ACM Student Chapter

May 2021 - May 2022

- **Recognized:** ACM **Outstanding** Chapter Award (2021-22), Best **Recruitment** Award **Globally** (2021-22)